CLASS 8 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 2

Chapter 8: **Nature of Matter: Elements, Compounds and Mixtures** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which of the following is brittle? 1

(a) copper **(b)** aluminium **(c)** gold **(d)** sulphur

2. Which is a homogeneous mixture? 1

(a) oil and water **(b)** iron filings and sulphur

(c) salt solution **(d)** sand and water

3. The symbol for potassium is: 1

(a) K **(b)** Po **(c)** Pt **(d)** P

4. Which element is a liquid non-metal at room temperature? 1

(a) mercury **(b)** chlorine **(c)** iodine **(d)** bromine

5. Which is a heterogeneous mixture? 1

(a) brass **(b)** sugar solution **(c)** sand and water **(d)** air

6. The symbol for iron is: 1

(a) F **(b)** Fe **(c)** Ir **(d)** I

7. Non-metals are generally: 1

(a) good conductors of heat **(b)** lustrous

(c) malleable **(d)** poor conductors of heat and electricity

8. Brass is an alloy of: 1

(a) copper and tin **(b)** copper and zinc **(c)** iron and carbon **(d)** lead and tin

9. The smallest particle of an element that can take part in a chemical reaction is the: 1

(a) compound **(b)** molecule **(c)** ion only **(d)** atom

10. Elements with properties of both metals and non-metals are called: 1

(a) metalloids **(b)** noble gases **(c)** alloys **(d)** compounds

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. What are symbols of elements? Write the symbols of carbon and calcium. 2

12. Name the elements present in 2

(a) water

(b) carbon dioxide.

13. Name two metals and two non-metals. 2

14. Differentiate between a homogeneous and a heterogeneous mixture with one example each. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Differentiate between a mixture and a compound in three points. 3

16. State three uses of non-metals in daily life. 3

17. Why are noble gases used in lighting? Name three of them. 3

OR

Describe an activity to show that heating iron and sulphur forms a compound. 3

18. State three uses of metals in daily life. 3

OR

State three tests/properties by which a metal can be distinguished from a non-metal. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain the activity of heating iron filings and sulphur:
(a) as a mixture
(b) as a compound. State differences. 5

OR

Discuss important properties of five elements and their uses. 5

20. Explain homogeneous and heterogeneous mixtures with three examples each and ways to separate them. 5

OR

Explain the law of constant proportions with an example of water. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND